APPENDIX B
CONVERGENCE ANALYSIS

This section provides the convergence analysis for our PiFLoRA framework, following in [82], [83]. We analyze the convergence of the local LoRA adapters under the iterative continual fine-tuning process, ensuring that the global model achieves a non-convex convergence rate.

Consider a set of N clients participating in federated fine-tuning with T communication rounds. Let η denote the learning rate, and let E denote the number of local iterations each client performs per communication round. In each round, the server broadcasts the global LoRA module $\mathcal{A}(\theta)$ (where $\mathcal{A}(\cdot)$ denotes the LoRA module structure and θ are the module parameters) of the selected K clients. Each client shares the same global LLM, and the personalization update is realized solely through the distinct LoRA module parameters θ_k. We define the loss function of client c_k at local iteration $e \in \{0, \dots, E\}$ within global round t as $\mathcal{L}_k^{t,e}(\theta_k^{t,e})$, and similarly denote the global LoRA-adapter model loss by $\mathcal{L}^t(\theta^t)$. To analyze convergence, we make the following standard assumptions:

Assumption 1 (L_1-Lipschitz smoothness)**.** *For each client $c_k \in \mathcal{C}$, the local loss function is L_1-Lipschitz smooth [83], [82], i.e.,*

$$\|\nabla\mathcal{L}_k^{t_1}(\theta_k^{t_1}; d_k) - \nabla\mathcal{L}_k^{t_2}(\theta_k^{t_2}; d_k)\| \le L_1\|\theta_k^{t_1} - \theta_k^{t_2}\|, \tag{18}$$

This also implies the following quadratic bound,

$$\mathcal{L}_k^{t_1} - \mathcal{L}_k^{t_2} \le \left\langle \nabla\mathcal{L}_k^{t_2}, (\theta_k^{t_1} - \theta_k^{t_2})\right\rangle + \frac{L_1}{2}\|\theta_k^{t_1} - \theta_k^{t_2}\|_2^2, \tag{19}$$
$$\forall t_1, t_2 \in \mathcal{T}, c_k \in \mathcal{C}, d_k \in \mathcal{D}_k$$

Assumption 2 (Unbiased gradient with bounded variance)**.** *Denoting $\gamma_k \in \mathcal{D}_k$ as a batch of client c_k's local data, the variance of the stochastic gradient of her local LoRA adapter $\nabla\mathcal{L}_k^t(\theta_k^t; \gamma_k^t)$ is unbiased, i.e.,*

$$\mathbb{E}_{\gamma_k \in \mathcal{D}_k}\left[\nabla\mathcal{L}_k^t(\theta_k^t; \gamma_k^t)\right] = \nabla\mathcal{L}_k^t(\theta_k^t) \tag{20}$$

and bounded by:

$$\mathbb{E}_{\gamma_k \in \mathcal{D}_k}\left[\|\nabla\mathcal{L}_k^t(\theta_k^t; \gamma_k^t) - \nabla\mathcal{L}_k^t(\theta_k^t)\|_2^2\right] \le \sigma^2. \tag{21}$$

Leveraging these assumptions, we derive the following lemmas and theorems.

Lemma 1 (Bounded local loss)**.** *Given Assumption 1 and Assumption 2, for any client c_k and communication round t, the local loss after E iterations of local update can be bounded as follows:*

$$\mathbb{E}[\mathcal{L}_k^{t+1,0}] \le \mathcal{L}_k^{t,0} - \left(\eta - \frac{L_1\eta^2}{2}\right)\sum_{e=0}^{E-1}\|\nabla\mathcal{L}_k^{t,e}\|^2 + \frac{L_1 E\eta^2\sigma^2}{2} \tag{22}$$

Lemma 1 provides a deviation bound for the local objective after each communication round, ensuring convergence can be guaranteed when given a suitable learning rate η.

Proof. The local LoRA module is updated as:

$$\theta_k^{t,e+1} = \theta_k^{t,e} - \eta\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e+1}; \gamma_k^t) \tag{23}$$

where $\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)$ is the stochastic gradient computed on a batch γ_k^t. Leveraging Assumption 1, the round-wise change in loss function can be bounded in:

$$\mathcal{L}_k^{t,e+1} \le \mathcal{L}_k^{t,e} + \left\langle \nabla\mathcal{L}_k^{t,e}, \theta_k^{t,e+1} - \theta_k^{t,e}\right\rangle + \frac{L_1}{2}\left\|\theta_k^{t,e+1} - \theta_k^{t,e}\right\|_2^2 \tag{24}$$

Substitute the update rule Equation (23):

$$\theta_k^{t,e+1} - \theta_k^{t,e} = -\eta\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t) \tag{25}$$

Thus, it can be derived as:

$$\begin{aligned}\mathcal{L}_k^{t,e+1} \le& \mathcal{L}_k^{t,e} - \eta\left\langle \nabla\mathcal{L}_k^{t,e}, \nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right\rangle \\ &+ \frac{L_1\eta^2}{2}\left\|\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right\|_2^2\end{aligned} \tag{26}$$

Taking the expectation over the batch γ_k^t with the Assumption 2, and using Assumption 2 we have:

$$\mathbb{E}_{\gamma_k^t}\left[\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right] = \nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}) \tag{27}$$

Then the inner product becomes:

$$\mathbb{E}_{\gamma_k^t}\left[\left\langle \nabla\mathcal{L}_k^{t,e}, \nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right\rangle\right] = \left\langle \nabla\mathcal{L}_k^{t,e}, \nabla\mathcal{L}_k^{t,e}\right\rangle = \left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \tag{28}$$

For the quadratic term, it can be derived as:

$$\mathbb{E}_{\gamma_k^t}\left[\left\|\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right\|_2^2\right] = \mathbb{E}_{\gamma_k^t}\left[\left\|\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t) - \nabla\mathcal{L}_k^{t,e} + \nabla\mathcal{L}_k^{t,e}\right\|_2^2\right] \tag{29}$$

Expanding the norm:

$$\begin{aligned}\left\|\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right\|_2^2 =& \left\|\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t) - \nabla\mathcal{L}_k^{t,e}\right\|_2^2 + \left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \\ &+ 2\left\langle \nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t) - \nabla\mathcal{L}_k^{t,e}, \nabla\mathcal{L}_k^{t,e}\right\rangle\end{aligned} \tag{30}$$

Since the expectation of the cross term is zero (due to unbiasedness), and using the variance bound from Assumption 2:

$$\mathbb{E}_{\gamma_k^t}\left[\left\|\nabla\mathcal{L}_k^{t,e}(\theta_k^{t,e}; \gamma_k^t)\right\|_2^2\right] \le \sigma^2 + \left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \tag{31}$$

Thus, the expected local loss for LoRA module update is:

$$\mathbb{E}\left[\mathcal{L}_k^{t,e+1}\right] \le \mathcal{L}_k^{t,e} - \eta\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 + \frac{L_1\eta^2}{2}\left(\sigma^2 + \left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2\right) \tag{32}$$

Simplified as:

$$\mathbb{E}\left[\mathcal{L}_k^{t,e+1}\right] \le \mathcal{L}_k^{t,e} - \left(\eta - \frac{L_1\eta^2}{2}\right)\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 + \frac{L_1\eta^2\sigma^2}{2} \tag{33}$$

Summing over $e = 0$ to $E-1$, it completes the proof with:

$$\mathbb{E}\left[\mathcal{L}_k^{t+1,0}\right] \le \mathcal{L}_k^{t,0} - \left(\eta - \frac{L_1\eta^2}{2}\right)\sum_{e=0}^{E-1}\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 + \frac{L_1 E\eta^2\sigma^2}{2} \tag{34}$$

□

With Lemma 1 and the above prove, taking expectation over all clients and all communication rounds, we further derive Theorem 2 as follows:

Theorem 2 (Non-convexv convergence of PiFLoRA)**.** *Under Assumptions 1 and 2, the PiFLoRA framework achieves a non-convex convergence rate of $\mathcal{O}\left(\frac{1}{\mathcal{T}}\right)$ for federated fine-tuning over $\mathcal{T}$ communication rounds.*

Letting $\Delta = \mathcal{L}^{0,0} - \mathcal{L}^* > 0$ be the initial optimality loss gap, where $\mathcal{L}^{0,0} = \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_k^{0,0}$ is the initial average loss across clients and $\mathcal{L}^*$ is the optimal loss. The average squared gradient norm over $\mathcal{T}$ rounds, with E local iterations per round, satisfies:

$$\frac{1}{\mathcal{T}}\sum_{t=0}^{\mathcal{T}-1}\sum_{e=0}^{E-1}\frac{1}{K}\sum_{k=1}^{K}\|\nabla\mathcal{L}_k^{t,e}\|_2^2 \leq \frac{\Delta}{\mathcal{T}\left(\eta - \frac{L_1\eta^2}{2}\right)} + \frac{L_1 E\eta^2\sigma^2}{2\left(\eta - \frac{L_1\eta^2}{2}\right)} \tag{35}$$

To guarantee convergence to a stationary point with tolerance ϵ, the learning rate η must satisfy:

$$\eta < \frac{2\epsilon}{L_1(E\sigma^2 + \epsilon)} \tag{36}$$

Proof. From Lemma 1, we have Equation (22), rearranging to isolate the gradient norm term, we have:

$$\left(\eta - \frac{L_1\eta^2}{2}\right)\sum_{e=0}^{E-1}\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \leq \mathcal{L}_k^{t,0} - \mathbb{E}\left[\mathcal{L}_k^{t+1,0}\right] + \frac{L_1 E\eta^2\sigma^2}{2} \tag{37}$$

Summing over all communication rounds $t = 0$ to $\mathcal{T} - 1$:

$$\left(\eta - \frac{L_1\eta^2}{2}\right)\sum_{t=0}^{\mathcal{T}-1}\sum_{e=0}^{E-1}\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \leq \sum_{t=0}^{\mathcal{T}-1}\left(\mathcal{L}_k^{t,0} - \mathbb{E}\left[\mathcal{L}_k^{t+1,0}\right]\right) + \frac{\mathcal{T}L_1 E\eta^2\sigma^2}{2} \tag{38}$$

Since $\mathcal{L}_k^{T,0} \geq \mathcal{L}^*$ where $\mathcal{L}^*$ denotes the optimal loss, we can derive: $\mathcal{L}_k^{0,0} - \mathbb{E}\left[\mathcal{L}_k^{\mathcal{T},0}\right] \leq \mathcal{L}_k^{0,0} - \mathcal{L}^*$. And the sum of the loss differences telescopes:

$$\begin{aligned}\sum_{t=0}^{\mathcal{T}-1}\left(\mathcal{L}_k^{t,0} - \mathbb{E}\left[\mathcal{L}_k^{t+1,0}\right]\right) &= \mathcal{L}_k^{0,0} - \mathbb{E}\left[\mathcal{L}_k^{\mathcal{T},0}\right] \\ &\leq \mathcal{L}_k^{0,0} - \mathcal{L}^* = \Delta,\end{aligned} \tag{39}$$

Thus, it can be further derived as:

$$\left(\eta - \frac{L_1\eta^2}{2}\right)\sum_{t=0}^{\mathcal{T}-1}\sum_{e=0}^{E-1}\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \leq \Delta + \frac{\mathcal{T}L_1 E\eta^2\sigma^2}{2} \tag{40}$$

Averaging over all clients $c_k \in \mathcal{C}$ (with K clients) and dividing by $\mathcal{T}\left(\eta - \frac{L_1\eta^2}{2}\right)$:

$$\frac{1}{\mathcal{T}}\sum_{t=0}^{\mathcal{T}-1}\sum_{e=0}^{E-1}\frac{1}{K}\sum_{k=1}^{K}\left\|\nabla\mathcal{L}_k^{t,e}\right\|_2^2 \leq \frac{\Delta}{\mathcal{T}\left(\eta - \frac{L_1\eta^2}{2}\right)} + \frac{L_1 E\eta^2\sigma^2}{2\left(\eta - \frac{L_1\eta^2}{2}\right)} \tag{41}$$

To ensure the average gradient norm is below a tolerance ϵ as $\mathcal{T} \to \infty$, the variance term must satisfy:

$$\frac{L_1 E\eta^2\sigma^2}{2\left(\eta - \frac{L_1\eta^2}{2}\right)} < \epsilon \tag{42}$$

Solving the inequality, we can get:

$$\begin{aligned}L_1 E\eta^2\sigma^2 < 2\epsilon\left(\eta - \frac{L_1\eta^2}{2}\right) &\implies L_1 E\eta^2\sigma^2 + \epsilon L_1\eta^2 < 2\epsilon\eta \\ &\implies \eta(L_1 E\sigma^2 + \epsilon L_1) < 2\epsilon \\ &\implies \eta < \frac{2\epsilon}{L_1(E\sigma^2 + \epsilon)}\end{aligned} \tag{43}$$

Since $E \geq 1$, this bound ensures a suitable learning rate η exists to achieve the desired tolerance. This completes the proof. □

Under the specified conditions, our PiFLoRA framework achieves a non-convex convergence rate of $\mathcal{O}(\frac{1}{\mathcal{T}})$ that follows from the $\frac{\Delta}{\mathcal{T}}$ term dominating as $\mathcal{T}$ increases, demonstrating rigorous theoretical guarantees for practical performance.